Analytics Vidhya

Optimize and Improve Prompts Automatically



Instructor

Dipanjan Sarkar

Head of Community & Principal AI Scientist at Analytics Vidhya

Google Developer Expert - ML & Cloud Champion Innovator

Published Author

Why use the prompt improver?

The prompt improver quickly enhances prompts through automated analysis. It excels at making prompts more robust for complex tasks that require high accuracy.

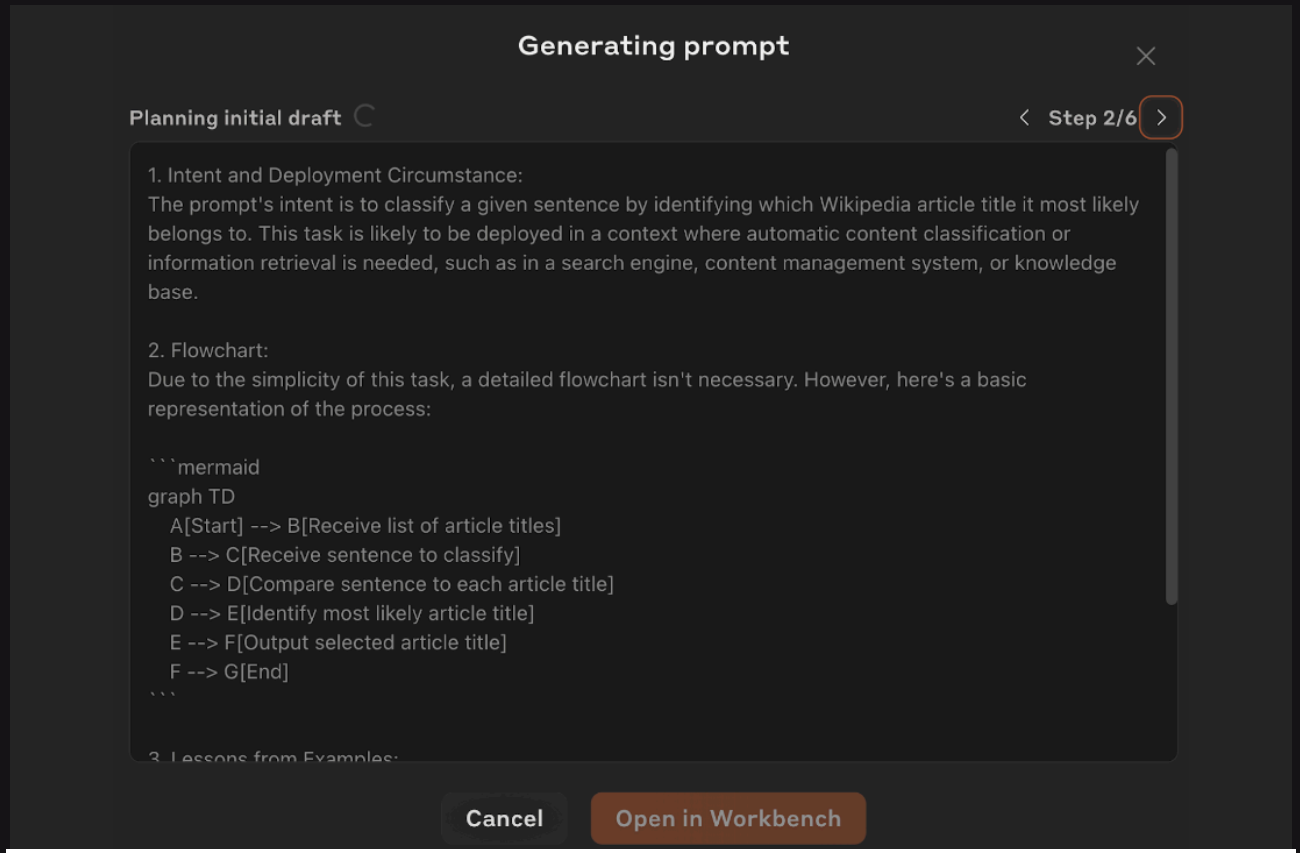
Analytics Vidhya

Prompt Improver

To optimize your prompts with Claude

Before you begin:

- A Prompt Template to improve
- Feedback on current issues with Claude's outputs (optional but recommended)
- Example inputs and ideal outputs (optional but recommended)



YOU CAN WATCH THESE STEPS HAPPEN IN REAL-TIME IN THE IMPROVEMENT MODAL.

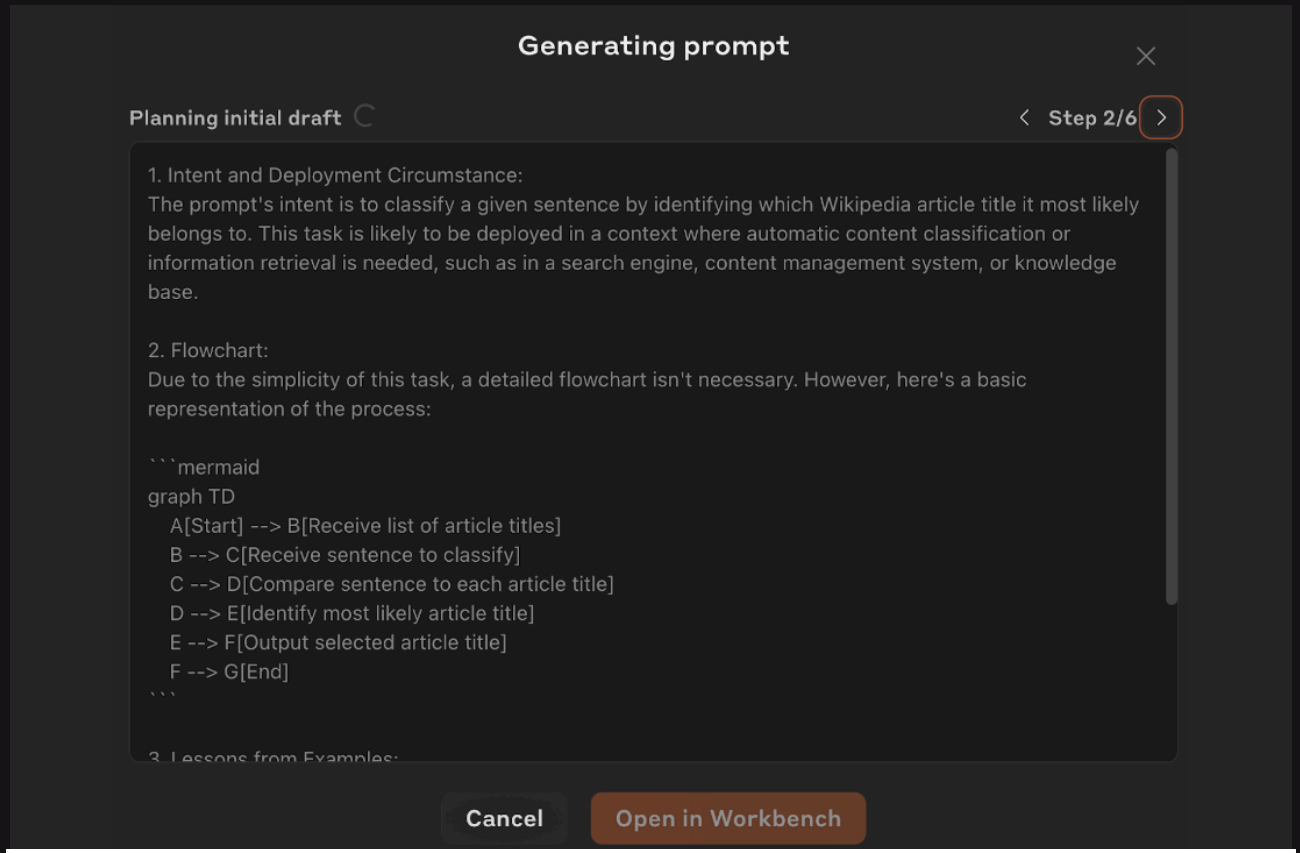
Analytics Vidhya

Prompt Improver

How the prompt improver works

The prompt improver enhances your prompts in 4 steps:

- **Example identification**: Locate and extract examples from your prompt template
- **Initial draft**: Creates a structured template with clear sections and XML tags
- **Chain of thought refinement**: Adds and refines detailed reasoning instructions
- **Example enhancement**: Updates examples to demonstrate the new reasoning process



YOU CAN WATCH THESE STEPS HAPPEN IN REAL-TIME IN THE IMPROVEMENT MODAL.

Analytics Vidhya

Original prompt

```
From the following list of Wikipedia article titles, identify which arti
Respond with just the article title and nothing else.

Article titles:
{{titles}}

Sentence to classify:
{{sentence}}
```

Improved prompt

```
You are an intelligent text classification system specialized in matchin

First, review the following list of Wikipedia article titles:
<article_titles>
{{titles}}
</article_titles>

Now, consider this sentence that needs to be classified:
<sentence_to_classify>
{{sentence}}
</sentence_to_classify>

Your goal is to determine which article title from the provided list best mat

1. List the key concepts from the sentence
2. Compare each key concept with the article titles
3. Rank the top 3 most relevant titles and explain why they are relevant
4. Select the most appropriate article title that best encompasses or relates

Wrap your analysis in <analysis> tags. Include the following:
- List of key concepts from the sentence
- Comparison of each key concept with the article titles
- Ranking of top 3 most relevant titles with explanations
- Your final choice and reasoning

After your analysis, provide your final answer: the single most appropriate Wi

Output only the chosen article title, without any additional text or explanati
```

Prompt Improver

To optimize your prompts with Claude

The prompt improver generates templates with:

- Detailed chain-of-thought instructions that guide Claude's reasoning process and typically improve its performance
- Clear organization using XML tags to separate different components
- Standardized example formatting that demonstrates step-by-step reasoning from input to output
- Strategic prefills that guide Claude's initial responses

Analytics Vidhya

Original prompt

```
From the following list of Wikipedia article titles, identify which arti th
Respond with just the article title and nothing else.

Article titles:
{{titles}}

Sentence to classify:
{{sentence}}
```

Improved prompt

```
You are an intelligent text classification system specialized in matchin ent

First, review the following list of Wikipedia article titles:
<article_titles>
{{titles}}
</article_titles>

Now, consider this sentence that needs to be classified:
<sentence_to_classify>
{{sentence}}
</sentence_to_classify>

Your goal is to determine which article title from the provided list best mat

1. List the key concepts from the sentence
2. Compare each key concept with the article titles
3. Rank the top 3 most relevant titles and explain why they are relevant
4. Select the most appropriate article title that best encompasses or relates

Wrap your analysis in <analysis> tags. Include the following:
- List of key concepts from the sentence
- Comparison of each key concept with the article titles
- Ranking of top 3 most relevant titles with explanations
- Your final choice and reasoning

After your analysis, provide your final answer: the single most appropriate Wi

Output only the chosen article title, without any additional text or explanati
```

Prompt Improver

To optimize your prompts with Claude

The prompt improver works best for:

- Complex tasks requiring detailed reasoning
- Situations where accuracy is more important than speed
- Problems where Claude's current outputs need significant improvement

Analytics Vidhya

▾ Original prompt

```
From the following list of Wikipedia article titles, identify which arti th
Respond with just the article title and nothing else.

Article titles:
{{titles}}

Sentence to classify:
{{sentence}}
```

▾ Improved prompt

```
You are an intelligent text classification system specialized in matchin ent

First, review the following list of Wikipedia article titles:
<article_titles>
{{titles}}
</article_titles>

Now, consider this sentence that needs to be classified:
<sentence_to_classify>
{{sentence}}
</sentence_to_classify>

Your goal is to determine which article title from the provided list best mat

1. List the key concepts from the sentence
2. Compare each key concept with the article titles
3. Rank the top 3 most relevant titles and explain why they are relevant
4. Select the most appropriate article title that best encompasses or relates

Wrap your analysis in <analysis> tags. Include the following:
- List of key concepts from the sentence
- Comparison of each key concept with the article titles
- Ranking of top 3 most relevant titles with explanations
- Your final choice and reasoning

After your analysis, provide your final answer: the single most appropriate Wi

Output only the chosen article title, without any additional text or explanati
```

Prompt Improver

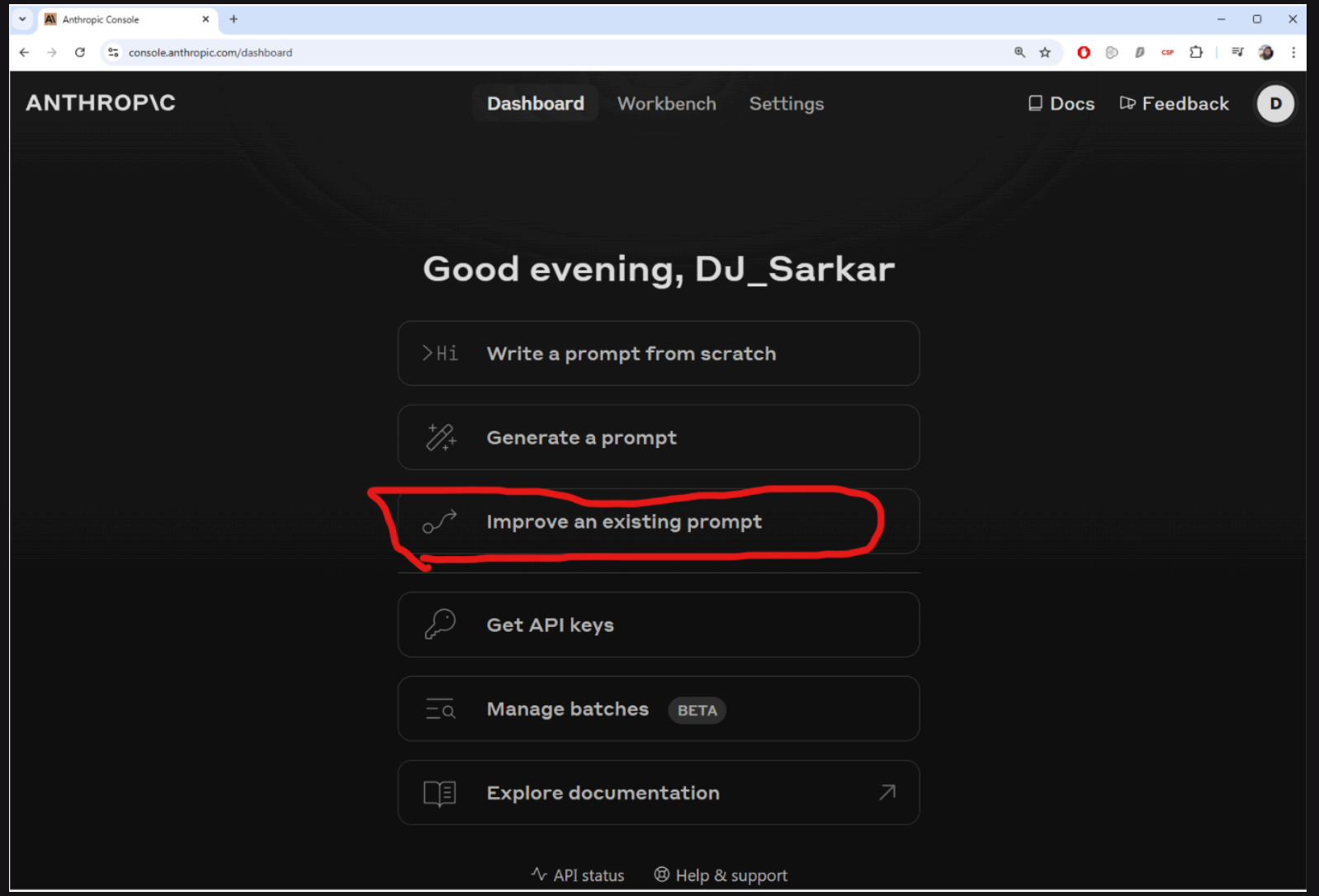
To optimize your prompts with Claude

Typically, the improved prompt:

- Adds clear step-by-step reasoning instructions
- Uses XML tags to organize content
- Provides explicit output formatting requirements
- Guides Claude through the analysis process

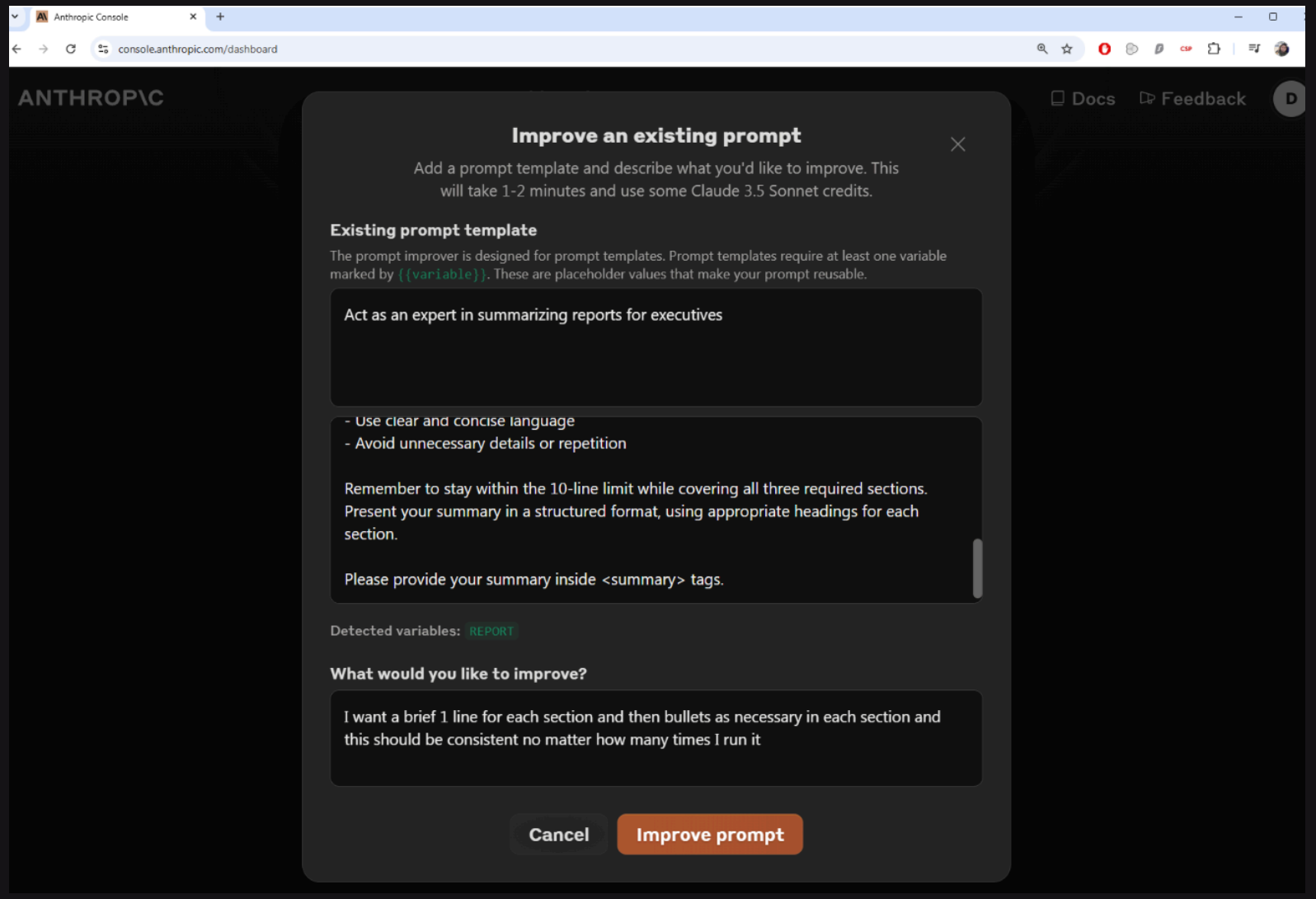
Analytics Vidhya

Use Claude to Improve Existing Prompts Faster



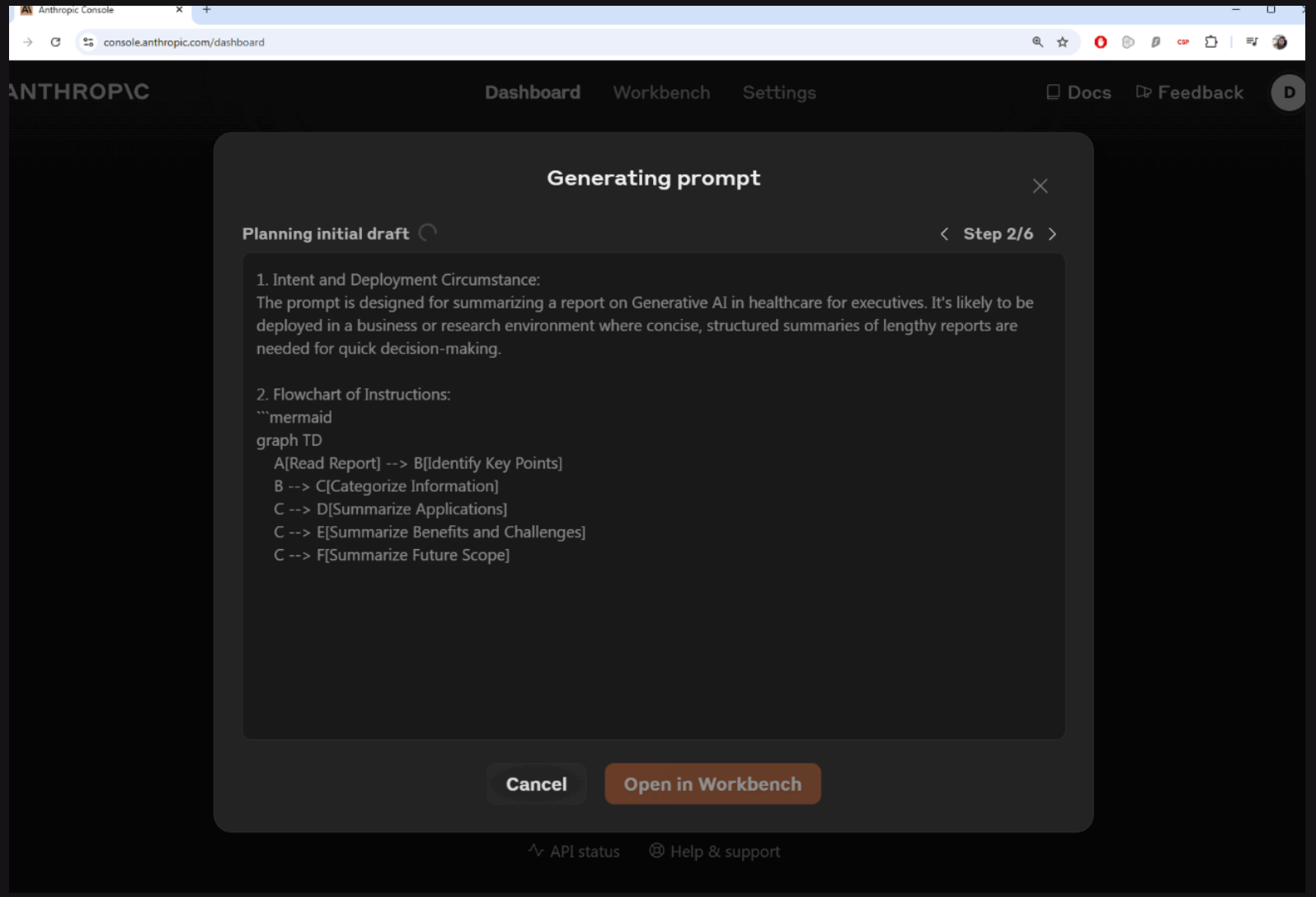
Analytics Vidhya

Use Claude to Improve Existing Prompts Faster



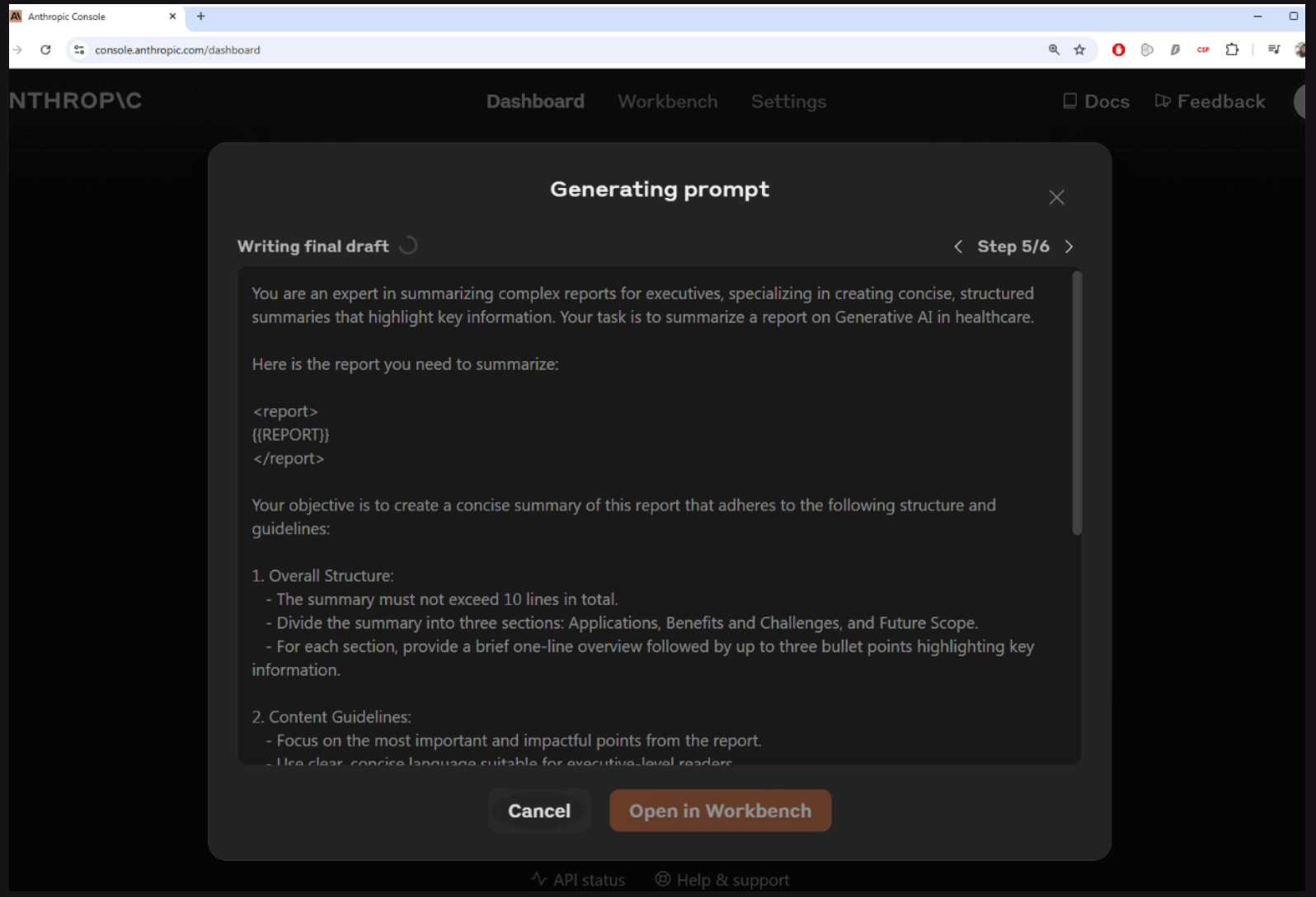
Analytics Vidhya

Use Claude to Improve Existing Prompts Faster



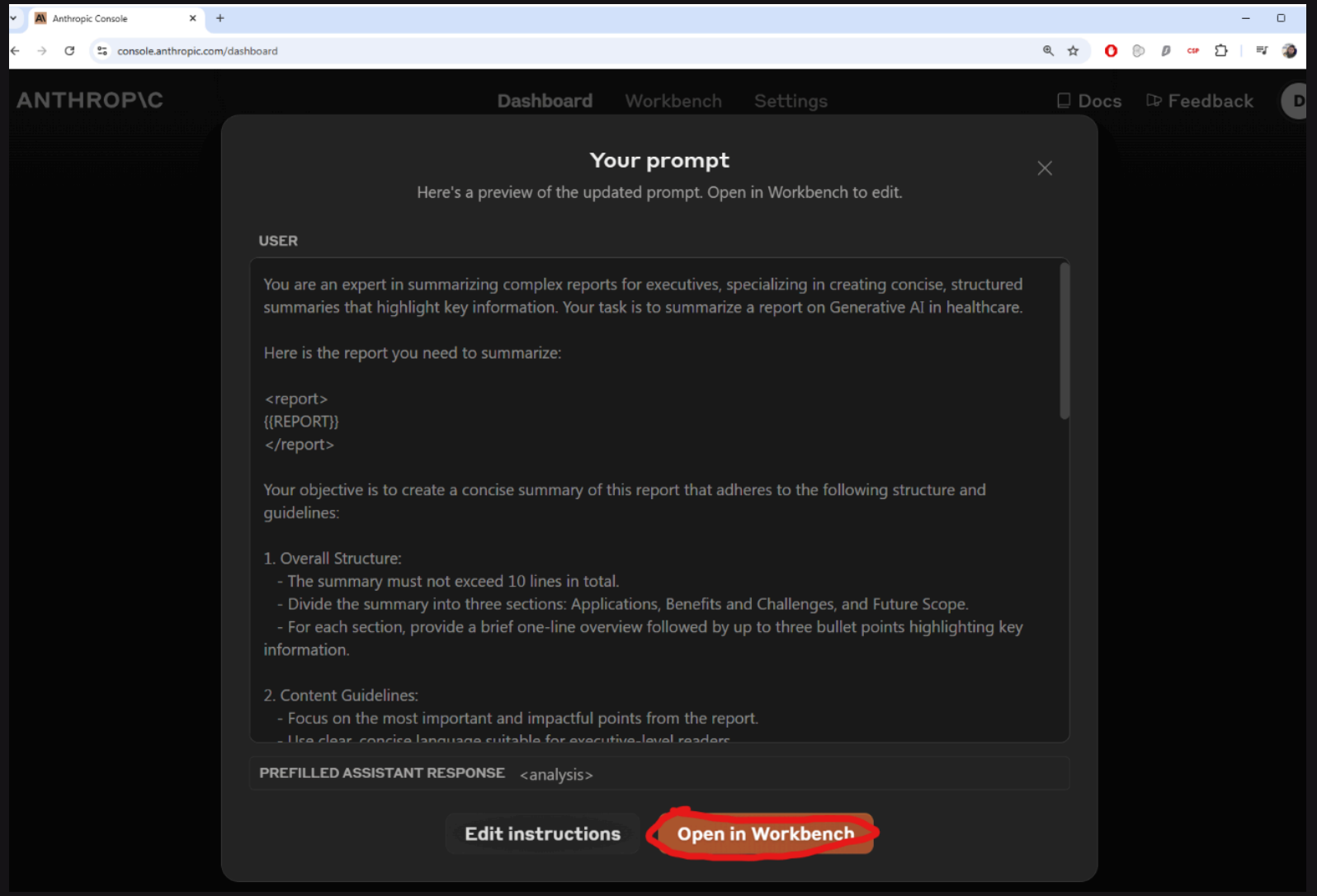
Analytics Vidhya

Use Claude to Improve Existing Prompts Faster



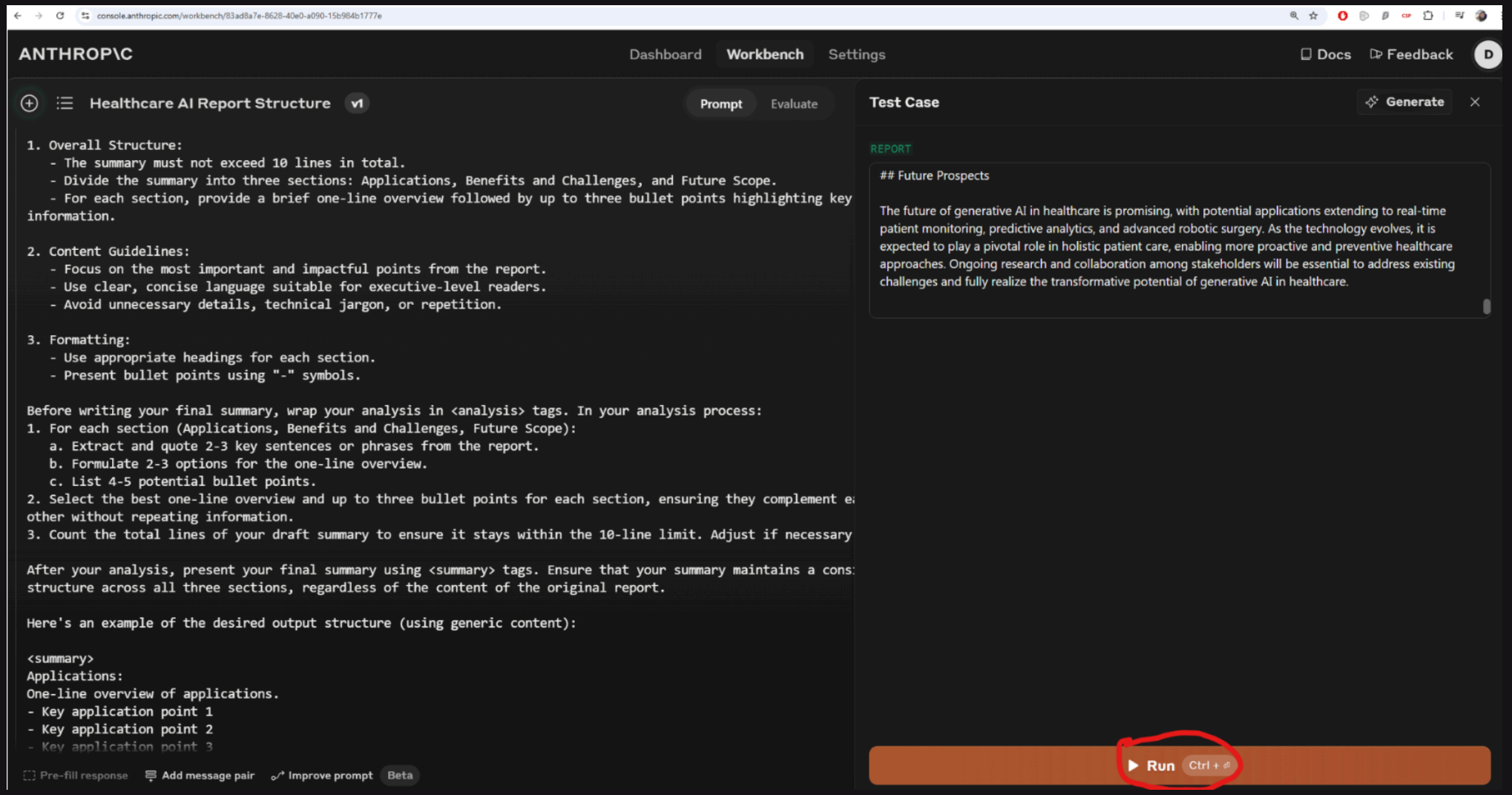
Analytics Vidhya

Use Claude to Improve Existing Prompts Faster



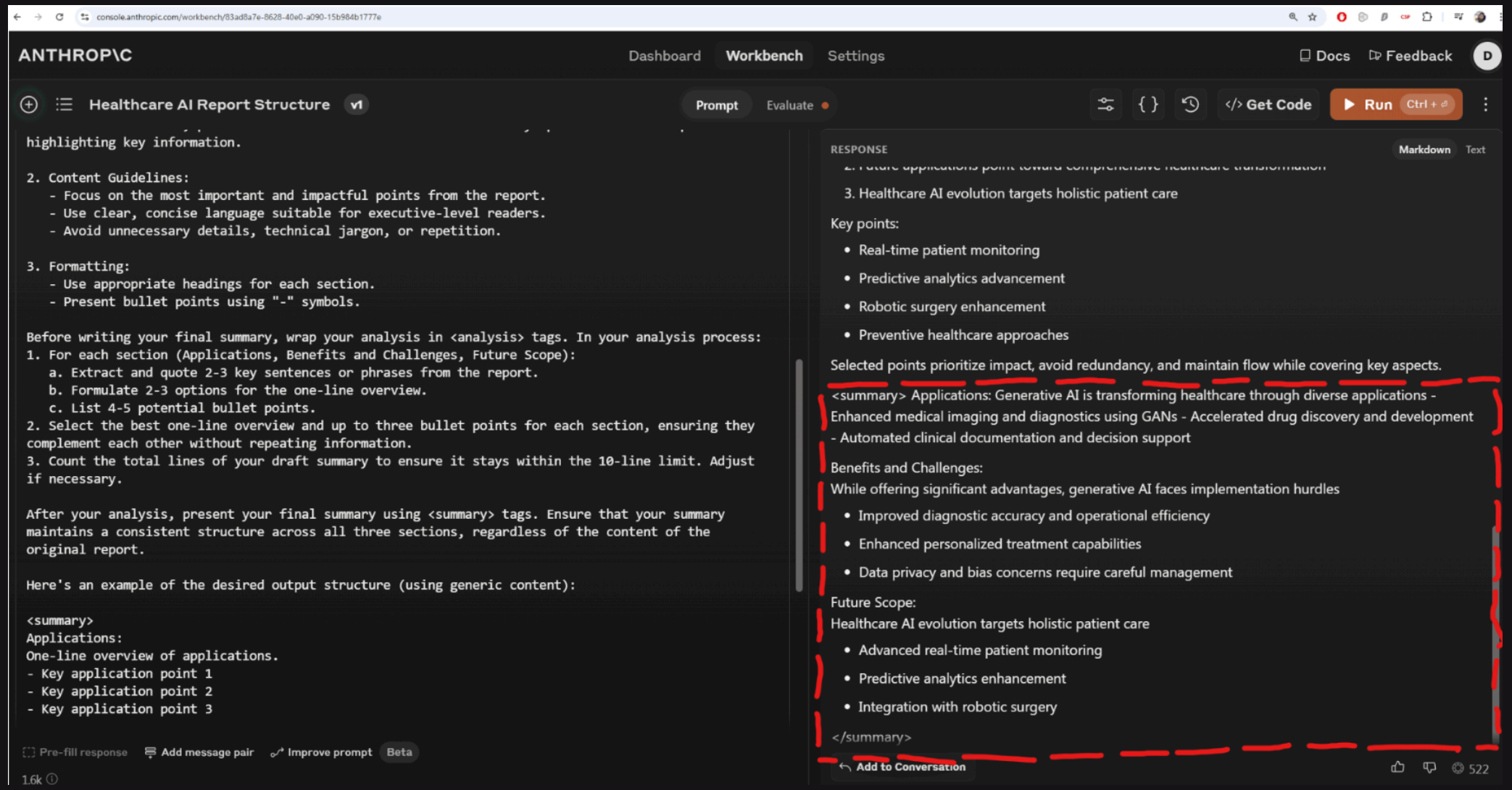
Analytics Vidhya

Use Claude to Improve Existing Prompts Faster



Analytics Vidhya

Use Claude to Improve Existing Prompts Faster



Analytics Vidhya

Thank You

Analytics Vidhya